UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 31/01/2019*

EXERCISE

We observe N samples (for n=0, 1,…, N-1, with N>>1) of a realization of the random process $X[n]=s[n]+W[n]$, where $s[n]=A\cos\left(2\pi F_0 n\right)$ is the signal of interest (SOI), A is the amplitude of the SOI modelled as an unknown deterministic parameter, and $W[n]$ is a zero-mean white Gaussian process with unknown power σ_W^2.

(1) Derive the signal to noise ratio (SNR) $\gamma \triangleq E\left\{\|\mathbf{s}\|^2\right\} \Big/ E\left\{\|\mathbf{W}\|^2\right\}$, where **s** and **W** are the vectors containing the N samples of signal and noise, respectively. Then, derive the log-likelihood function $LLF(A,\sigma_W^2)=\ln f_X(\mathbf{x};A,\sigma_W^2)$, where $f_X(\mathbf{x};A,\sigma_W^2)$ is the probability density function (PDF) of the observed data vector $\mathbf{X}=\left[X[0] \quad X[1] \quad \cdots \quad X[N-1]\right]^T$.

(2) Derive the joint maximum likelihood (ML) estimators of the signal amplitude A and of the noise power σ_W^2.

(3) Derive the bias and the mean square error (MSE) of the ML estimator of A, establish if it is consistent or not, plot the MSE as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_W^2=1$.

(4) Derive the bias of the ML estimator of σ_W^2 and plot it as a function of the sample size N.

(5) Find the Cramér-Rao bounds (CRBs) on the joint estimation of A and σ_W^2 and establish if the ML estimator of A is efficient or not.

Solution.

(1) Derive the SNR and the log-likelihood function.

$E\left\{\|\mathbf{s}\|^2\right\} = \|\mathbf{s}\|^2 = \frac{A^2 N}{2}$, $E\left\{\|\mathbf{W}\|^2\right\} = N\sigma_W^2$, quindi $\gamma \triangleq E\left\{\|\mathbf{s}\|^2\right\} \Big/ E\left\{\|\mathbf{W}\|^2\right\} = \frac{A^2}{2\sigma_W^2}$

$X[n] \in N\left(A\cos\left(2\pi F_0 n\right), \sigma_W^2\right)$, $\left\{X[n]\right\}_{n=0}^{N-1}$ indipendent $\Rightarrow$ $\mathbf{X} \in N\left(A\mathbf{p}, \sigma_W^2 \mathbf{I}\right)$

where $\left[\mathbf{p}\right]_n = \cos\left(2\pi F_0 n\right)$ is perfectly known and $\|\mathbf{p}\|^2 = \frac{N}{2}$.

PDF in scalar format:

$$f_X(\mathbf{x}; A, \sigma_W^2) = \prod_{n=0}^{N-1} f_X(x[n]; A, \sigma_W^2) = \left(\frac{1}{\sqrt{2\pi\sigma_W^2}}\right)^N \exp\left(-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x[n] - A\cdot\cos\left(2\pi F_0 n\right)\right)^2\right)$$

PDF in vector format:

$$f_X(\mathbf{x}; A, \sigma_W^2) = \frac{1}{\left(2\pi\right)^{N/2}\sqrt{\left|\sigma_W^2\mathbf{I}\right|}} \exp\left(-\frac{1}{2}\left(\mathbf{x} - A\mathbf{p}\right)^T \left(\sigma_W^2 \mathbf{I}\right)^{-1}\left(\mathbf{x} - A\mathbf{p}\right)\right)$$

$$= \frac{1}{\left(2\pi\right)^{N/2}\left(\sigma_W^2\right)^{N/2}} \exp\left(-\frac{1}{2\sigma_W^2}\|\mathbf{x} - A\mathbf{p}\|^2\right)$$

The log-likelihood function is given by:

$$LLF(A, \sigma_W^2) = \ln f_X(\mathbf{x}; A, \sigma_W^2) = -\frac{N}{2}\ln\left(2\pi\right) - \frac{N}{2}\ln\left(\sigma_W^2\right) - \frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(x[n] - A\cdot\cos\left(2\pi F_0 n\right)\right)^2$$

$$= -\frac{N}{2}\ln\left(2\pi\right) - \frac{N}{2}\ln\left(\sigma_W^2\right) - \frac{1}{2\sigma_W^2}\|\mathbf{x} - A\mathbf{p}\|^2$$

(2) Derive the joint ML estimator of the unknown signal A and unknown noise power σ_W^2.

$$\frac{d\ln f_X(\mathbf{x}; A, \sigma_W^2)}{dA} = \frac{1}{\sigma_W^2}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\left(x[n] - A\cdot\cos\left(2\pi F_0 n\right)\right) = 0$$

$$\Rightarrow \quad \hat{A}_{ML} = \frac{\sum_{n=0}^{N-1} x[n]\cos\left(2\pi F_0 n\right)}{\sum_{n=0}^{N-1}\cos^2\left(2\pi F_0 n\right)} = \frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos\left(2\pi F_0 n\right) = \frac{2}{N}\mathbf{p}^T\mathbf{x}$$

$$\frac{d\ln f_X(\mathbf{x};A,\sigma_W^2)}{d\sigma_W^2}=-\frac{N}{2\sigma_W^2}+\frac{1}{2\sigma_W^4}\sum_{n=0}^{N-1}\left(x[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)^2=0$$

$$\Rightarrow \quad \hat{\sigma}_{W,ML}^2=\frac{1}{N}\sum_{n=0}^{N-1}\left(x[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)^2$$

Putting together the two equations:

$$\hat{\sigma}_{W,ML}^2=\frac{1}{N}\sum_{n=0}^{N-1}\left(x[n]-\hat{A}_{ML}\cdot\cos\left(2\pi F_0 n\right)\right)^2=\frac{1}{N}\sum_{n=0}^{N-1}\left(x[n]-\frac{2}{N}\sum_{k=0}^{N-1}x[k]\cos\left(2\pi F_0 k\right)\cdot\cos\left(2\pi F_0 n\right)\right)^2$$

(3) Derive the bias and the mean square error (MSE) of the ML estimator of *A*, establish if it is consistent or not, plot the MSE as a function of γ and find the minimum value of *N* which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_W^2=1$.

The ML estimate of A does not require knowledge of the noise power. Moreover, since it is a linear estimator and the observed data are Gaussian distributed, the estimate is also Gaussian distributed.

$$E\left\{\hat{A}_{ML}\right\}=E\left\{\frac{2}{N}\mathbf{p}^T\mathbf{x}\right\}=\frac{2}{N}\mathbf{p}^T E\{\mathbf{x}\}=\frac{2}{N}\mathbf{p}^T\left(A\mathbf{p}\right)=A$$

Hence the estmator is unbiased.

$$MSE\left\{\hat{A}_{ML}\right\}=\mathrm{var}\left\{\hat{A}_{ML}\right\}=\mathrm{var}\left\{\frac{2}{N}\mathbf{p}^T\mathbf{x}\right\}=E\left\{\frac{2}{N}\mathbf{p}^T\mathbf{w}\left(\frac{2}{N}\mathbf{p}^T\mathbf{w}\right)^T\right\}=\frac{4}{N^2}\mathbf{p}^T\left\{\mathbf{w}\mathbf{w}^T\right\}\mathbf{p}=\frac{2\sigma_W^2}{N}$$

The estimator is consistent.

$$MSE\left\{\hat{A}_{ML}\right\}=\frac{2\sigma_W^2}{N}=\frac{2}{N}\le 10^{-3}\quad\Rightarrow\quad N\ge 2000$$

(4) Derive the bias of the ML estimator of σ_W^2 and plot it as a function of the sample size *N*.

$$E\left\{\hat{\sigma}_{W,ML}^2\right\}=\frac{1}{N}\sum_{n=0}^{N-1}E\left\{\left(X[n]-\hat{A}_{ML}\cdot\cos\left(2\pi F_0 n\right)\right)^2\right\}$$

$$E\left\{\left(X[n]-\hat{A}_{ML}\cdot\cos\left(2\pi F_0 n\right)\right)^2\right\}=E\left\{X^2[n]\right\}+E\left\{\left(\hat{A}_{ML}\right)^2\right\}\cos^2\left(2\pi F_0 n\right)-2\cos\left(2\pi F_0 n\right)E\left\{X[n]\hat{A}_{ML}\right\}$$

$$E\left\{X^2[n]\right\} = \left(E\left\{X[n]\right\}\right)^2 + \mathrm{var}\left\{X[n]\right\} = A^2\cos^2\left(2\pi F_0 n\right) + \sigma_W^2$$

$$E\left\{\left(\hat{A}_{ML}\right)^2\right\} = \left(E\left\{\hat{A}_{ML}\right\}\right)^2 + \mathrm{var}\left\{\hat{A}_{ML}\right\} = A^2 + \frac{2\sigma_W^2}{N}$$

$$\begin{aligned}
E\left\{X[n]\hat{A}_{ML}\right\} &= E\left\{X[n]\hat{A}_{ML}\right\} = E\left\{X[n]\cdot\frac{2}{N}\sum_{k=0}^{N-1}X[k]\cos\left(2\pi F_0 k\right)\right\} \\
&= \frac{2}{N}\sum_{k=0}^{N-1}E\left\{X[n]X[k]\right\}\cos\left(2\pi F_0 k\right) \\
&= \frac{2}{N}\sum_{k=0}^{N-1}\left(A^2\cos\left(2\pi F_0 n\right)\cos\left(2\pi F_0 k\right) + \sigma_W^2\delta[k-n]\right)\cos\left(2\pi F_0 k\right) \\
&= A^2\cos\left(2\pi F_0 n\right) + \frac{\sigma_W^2}{N}\cos\left(2\pi F_0 n\right) = \left(A^2 + \frac{2\sigma_W^2}{N}\right)\cos\left(2\pi F_0 n\right)
\end{aligned}$$

$$\begin{aligned}
E\left\{\hat{\sigma}_{W,ML}^2\right\} &= \frac{1}{N}\sum_{n=0}^{N-1}\left(A^2\cos^2\left(2\pi F_0 n\right) + \sigma_W^2\right) + \frac{1}{N}\sum_{n=0}^{N-1}\left(A^2 + \frac{2\sigma_W^2}{N}\right)\cos^2\left(2\pi F_0 n\right) \\
&\quad - \frac{2}{N}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\cdot\left(A^2 + \frac{2\sigma_W^2}{N}\right)\cos\left(2\pi F_0 n\right) \\
&= \frac{A^2}{2} + \sigma_W^2 + \frac{A^2}{2} + \frac{\sigma_W^2}{N} - \left(A^2 + \frac{2\sigma_W^2}{N}\right) = \sigma_W^2 - \frac{\sigma_W^2}{N} = \sigma_W^2\left(1 - \frac{1}{N}\right) \neq \sigma_W^2
\end{aligned}$$

$$bias\left\{\hat{\sigma}_{W,ML}^2\right\} = E\left\{\sigma_W^2 - \hat{\sigma}_{ML}^2\right\} = \frac{\sigma_W^2}{N}$$

Hence, the estimator of the noise power is biased but asymptotically unbiased.

(5) Find the Cramér-Rao bounds (CRBs) on the estimation of A and σ_W^2 and establish if the ML estimator of A is efficient or not.

$$\frac{d^2\ln f_X(\mathbf{x})}{dA^2} = \frac{d}{dA}\left(\frac{1}{\sigma_W^2}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\left(X[n] - A\cdot\cos\left(2\pi F_0 n\right)\right)\right) = -\frac{1}{\sigma_W^2}\sum_{n=0}^{N-1}\cos^2\left(2\pi F_0 n\right) = -\frac{N}{2\sigma_W^2}$$

$$-E\left\{\frac{d^2\ln f_X(\mathbf{x})}{dA^2}\right\} = \frac{N}{2\sigma_W^2}$$

$$\frac{d^2 \ln f_X(\mathbf{x})}{dAd\sigma_W^2} = \frac{d}{d\sigma_W^2}\left(\frac{1}{\sigma_W^2}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\left(X[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)\right)$$

$$= \frac{1}{\sigma_W^4}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)\left(X[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)$$

$$-E\left\{\frac{d^2 \ln f_X(\mathbf{x})}{dAd\sigma_W^2}\right\} = \frac{1}{\sigma_W^4}\sum_{n=0}^{N-1}\cos\left(2\pi F_0 n\right)E\left\{\left(X[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)\right\} = 0$$

$$\frac{d\ln f_X(\mathbf{x};A,\sigma_W^2)}{d\sigma_W^4} = \frac{d}{d\sigma_W^2}\left(-\frac{N}{2\sigma_W^2}+\frac{1}{2\sigma_W^4}\sum_{n=0}^{N-1}\left(X[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)^2\right)$$

$$= \frac{N}{2\sigma_W^4}-\frac{1}{\sigma_W^6}\sum_{n=0}^{N-1}\left(X[n]-A\cdot\cos\left(2\pi F_0 n\right)\right)^2$$

$$-E\left\{\frac{d\ln f_X(\mathbf{x};A,\sigma_W^2)}{d\sigma_W^4}\right\} = -\frac{N}{2\sigma_W^4}+\frac{N}{\sigma_W^6}\sigma_W^2 = \frac{N}{2\sigma_W^4}$$

Since the Fisher Information Matrix (FIM) is diagonal, we have:

$$CRB(A) = \frac{1}{-E\left\{\frac{d^2 \ln f_X(\mathbf{x})}{dA^2}\right\}} = \frac{2\sigma_W^2}{N} = MSE\left\{\hat{A}_{ML}\right\}$$

Hence, the ML estimator of A is efficient.

$$CRB(\sigma_W^2) = \frac{1}{-E\left\{\frac{d^2 \ln f_X(\mathbf{x})}{d\sigma_W^4}\right\}} = \frac{2\sigma_W^4}{N}$$

$\hat{\sigma}_{W,ML}^2$ is biased, so it cannot be efficient. It is asymptotically efficient, hence the asymptotical MSE is $MSE\left\{\hat{\sigma}_{W,ML}^2\right\} \overset{N\gg 1}{=} \frac{2\sigma_W^4}{N}$.